GCE

Rewarding Learning

Chief Examiner's and Principal Moderator's Report

Digital Technology



Summer Series 2025

Foreword

This booklet outlines the performance of candidates in all aspects of this specification for the Summer 2025 series.

CCEA hopes that the Chief Examiner's and/or Principal Moderator's report(s) will be viewed as a helpful and constructive medium to further support teachers and the learning process.

This booklet forms part of the suite of support materials for the specification. Further materials are available from the specification's microsite on our website at www.ccea.org.uk.

Contents

Assessment Unit AS 1	Approaches to Systems Development	3
Assessment Unit AS 2	Fundamentals of Digital Technology	6
Assessment Unit A2 1	Information Systems	10
Assessment Unit A2 2	Application Development (Case Study)	13
Contact details		15

GCE DIGITAL TECHNOLOGY

Chief Examiner's Report

Subject Overview

As the assessment of this specification moves on from the recent external challenges which impacted on all specifications, it is important to stress the evidence from this series of the excellent work of centres in preparing and supporting candidates for all modules. An increasing number of centres had properly prepared their candidates for this specification, and had taken on board the feedback in the previous Chief Examiner Reports.

The performance of candidates on the individual modules is described below. However, centres are urged to read previous Chief Examiner's Reports for GCE Digital Technology as there is still clear evidence that many candidates could benefit from a better appreciation of the challenges they face in trying to meet all the specification's assessment objectives, particularly AO3. Candidates must be familiar with Digital Technology terminology to properly understand the questions, and they must use appropriate Digital Technology terminology in their responses.

Assessment Unit AS 1 Approaches to Systems Development

Q1 **(a)** **(i)** Most candidates were familiar with the term 'user interface' and could provide the required detail.

(ii) This question proved to be much more discriminating than intended. Only a small number of candidates could identify the two elements as specified in the specification.

(b) **(i)** Most candidates were not properly familiar with source code and few complete answers were evident.

(ii) The overall level of response to this question was high. Although many candidates struggled to elaborate on the roles, most could identify at least two. A significant number of candidates referred to the writing of code which was given in the question stem.

(c) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate the use of observation and questionnaires as fact-finding methods when time was relatively short, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. Most candidates were familiar with both methods but many failed to provide the expected detail. As expected, very few candidates met the strict criteria for a top-band response.

Q2 **(a)** As in previous occasions when this topic was assessed, many candidates were confused by the difference between functional and nonfunctional requirements, referring instead to essential and non-essential requirements.

(b) The overall quality of the dataflow diagrams produced by candidates was disappointing. Many candidates ignored some of the three stages of the booking process as indicated by the bullet points in the question. Of those who did recognize the three stages, many were careless and/or inaccurate in the process names chosen – the process name should bear some relation to the process being performed. It is a continuing matter of concern that many candidates are not properly familiar with the dataflow diagram notations in the specification.

Q3 **(a)** Many candidates struggled to distinguish between the Design and Development stages in the creation of an information system. However, some complete answers were provided.

(b) The specification refers explicitly to the use of prototyping in the design of the user interface and the data model. Most candidates interpreted Part (i) and Part (ii) as the same question and proceeded to describe prototyping itself. Only a small number of candidates achieved any marks in these two question parts.

(c) **(i)** The overall level of response was very high. Most candidates could identify at least three components found in a test plan.

(ii) Acceptance testing was better understood than beta testing. Many candidates failed to distinguish between them and provided almost identical descriptions for both.

Q4 **(a)** The overall level of response to this question was very high. However, some candidates were penalised for providing one word responses such as ‘help’ for user documentation.

(b) Most candidates were properly familiar with the parallel changeover method. There were two significant weaknesses. Firstly, the tendency to use distinctly non-DT terminology such as ‘until the users are happy with it’ or that the old system was available ‘to fall back on’. Secondly, candidates who described the advantages and/or disadvantages of the changeover method.

(c) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate the role of adaptive maintenance and perfective maintenance in ensuring a system remains suitable for the user, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. Most candidates were familiar with both types of maintenance but many failed to provide the expected detail. As expected, very few candidates met the strict criteria for a top-band response.

Q5 **(a)** Only a small number of candidates were properly familiar with RAD and few complete answers were provided. Some candidates who demonstrated a familiarity with RAD failed to use the appropriate terminology, referring to subtasks or modules instead of builds, or time limits and deadlines instead of time frames.

(b) Most candidates were familiar with Gantt charts. Hence the overall level of response was very high. The only significant weakness of those candidates familiar with Gantt charts was a failure to make it clear that the components of the chart mostly refer to the stages of the project, not the project itself.

(c) In general, the use of CPA in project management was poorly understood. Few good quality answers were provided.

(d) Candidates were broadly familiar with the purpose of a disaster recovery plan. Most candidates could provide one relevant point.

(e) The overall level of response was good. Most candidates could identify as least two relevant points. However, candidates need to be reminded that very brief answers such as 'another location' are not acceptable at this level in any question. In this case, the nature or purpose of the alternative location must be stated in a DT context.

Q6 **(a)** Many candidates were not familiar with the two programming terms. Of the two terms, algorithm was the better understood – a number of candidates referred to syntax in the context of data.

(b) While many candidates were familiar to some extent with the two programming concepts, many struggled to explain them as required. Phrases such as 'yes/no variable' or 'a condition is met' are not appropriate DT terminology.

(c) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate the object-oriented approach with regard to code reuse, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. Most candidates were not properly familiar with the object-oriented approach. Many understood inheritance as simply copying and pasting code or 'taking on values'. As expected, very few candidates met the strict criteria for a top-band response.

Assessment Unit AS 2 Fundamentals of Digital Technology

Overview

This paper contained five, multi-part, questions. Marks available in various questions parts ranged from one mark to a maximum of nine marks covering all areas of the unit specification.

Evidence showed that all candidates had appropriate time to complete the paper. Questions were designed to ensure all candidates could engage meaningfully with the paper, achieving a level of success reflective of their knowledge and skills. The standard of answering varied. Some candidates exhibited confident, structured responses with clear reference to the question demands. These candidates demonstrated not only knowledge recall but also the ability to apply this knowledge critically. In contrast, some responses suggested misinterpretation or lack of preparation, particularly in questions requiring extended analysis or precision in technical terminology (candidates should be reminded of the importance of using technical terminology in all responses to assessment material at this level of assessment).

Some evidence existed which showed at times that candidates understood the concept being assessed but they provided insufficient detail in their responses to support the allocation of full marks in that particular question part. It is important that candidates take note of the marks being allocated to each question part so they can use this as an indication of the level of detail required for their response. It is also important to remind candidates that when responding to a QWC style question, they should attempt to provide detailed descriptions of all key terms referenced in the question stem before explaining/justifying/evaluating (as required by the QWC question). Not doing so can impact on their ability to access Band 2 in the QWC mark scheme.

Q1 **(a)** **(i)** Most candidates familiar with the concept of a character set and referenced the concept of it being a list of symbols, a smaller number were able to expand this answer to include reference to the idea that it was in binary format.

(ii) Most candidates were able to successfully identify in both decimal and in the form of 2n) how many different characters could be represented in binary using 7 bits.

(iii) A number of candidates were able to fully define the term 'control character'.

(b) Most candidates were able to successfully provide a 2's complement representation of the positive number provided in the question stem. There was some evidence in responses and working out that some candidates struggle with binary addition, this resulted in incorrect outcomes for some candidates when completing the negative number conversion to 2's complement format.

(c) Some candidates opted to represent 1gigabyte as 1000 megabytes as opposed to 1024 definition used in computer science and related fields. This was accepted by the marking team and will continue to be the case given that this is now a more widely accepted notation in the field.

(d) This was the first QWC question of the paper. This question was designed to be accessible and to encourage confidence in responding to this question style. Marks were allocated for level of detail in the description of both ASCII and Unicode. There was clear evidence that candidates understood both concepts well as the question was well answered by many. It is important to remind candidates however that when presented with two terms in a QWC style question, they should attempt to provide detailed descriptions of both terms before also then explaining/justifying/evaluating (as required by the QWC question). Not doing so can impact on their ability to access Band 2 in the QWC mark scheme.

Q2 **(a)** Most candidates were able to provide detailed definitions of 'Data', 'Information' and 'Knowledge'. Responses at times did not include suitable detailed examples from the stimulus provided in the question stem. Candidates should be reminded, at AS level, of the need to fully elaborate on the examples they provide in their response: for example, when providing an explanation of data, the example should state '500 is a string of characters' as opposed to 'e.g. 500' (which does not provide a context for the example).

(b) Most candidates were able to access at least two out of the four marks awarded in this question part. It was clear from candidate responses that most candidates know the features that can affect the quality of information. Some have difficulty providing suitable explanations of the features however.

(c) Most candidates were able to identify two methods of data verification although some did confuse verification with validation and provided examples of validation checks instead.

(d) Most candidates were aware of the general differences between hash and control totals. In some cases, the examples provided by candidates did not link to the scenario provided in the question stem.

Q3 **(a)** In providing definitions of 'Main Memory' some candidates incorrectly referenced ROM and Cache as being examples of main memory as opposed to considering the key features of main memory/RAM.

A significant number of candidates were able to provide detailed definitions of the terms 'Clock' and 'Port' but for some their definitions were somewhat generic and lacking in technical terminology, an important element of a detailed response at this level.

(b) Most candidates were able to correctly identify two internal buses found in a computer system.

(c) Candidates could in the main successfully explain why the PC was incremented during the fetch-execute cycle. Some candidates had difficulty explaining why data was loaded into the MDR during the fetch-execute cycle however. Where candidates provided a detailed technical definition of the MDR however, this was sufficient to ensure all marks for this question part were accessible.

(d) Many students struggled with this QWC question. This question was designed to act as a discriminatory question and as a result only a small number of candidates were awarded marks in Band 2 by. Where candidates scored highly in this question, they provided detailed, technical definitions of the processor types (in many cases using terminology referenced in the appropriate fact files), before evaluating each processor type regarding speed of processing.

Q4 **(a)** Most candidates were able to clearly identify four main features of WIMP user interfaces.

(b) While it was clear that many candidates were familiar with the main features of a command line interface a number of candidates approached the question from an incorrect perspective with some commenting on why the interface type was not appropriate for novice users. Candidates should be reminded of the importance of reading each question carefully to ensure perspectives are not confused.

(c) Candidates in general were able to provide some form of description of both bitmap and GIF image file formats though common misconceptions around compression was evident; most commonly that bitmap file types are compressed file types, and that GIF uses lossy compression; these misconceptions impacted on the marks awarded to a noticeable number of candidates.

(d) This QWC question proved accessible to many candidates. Most candidates were able to provide a reasonable definition of one or both software types referenced in the question stem. As stated previously, candidates should be reminded of the importance of fully defining all key terms referenced in QWC stems, in addition to ensuring an appropriate focus is placed on the evaluative element associated with the question. While many candidates were able to provide a good/detailed description of the software types, only a smaller proportion of the candidature included an appropriate focus on how the decision made could reduce costs for the business.

Q5 **(a)** This question was well answered by most candidates showing there is a clear understanding of the inputs and outputs associated with a typical ATM.

(b) While most candidates were able to provide detailed and appropriate definitions of URL, candidates seemed to have difficulty using appropriate terminology when describing web hosting.

(c) Candidate responses to this question were mixed with some not being awarded full marks due to the incorrect use of open and closing html tags. Where html tags are specifically referenced in the specification it would be advisable to candidates to learn their structures and purpose in preparation for such question types.

(d) This question was poorly answered by a large proportion of candidates showing an area of weakness in knowledge across the wider candidature. Some candidates who were awarded higher marks in this question were still not awarded full marks in the question as they were unable to identify all elements of the domain name or in some cases because they did not make appropriate reference to the example shown when providing their responses.

(e) In this final QWC question of the paper most candidates were able to give an appropriate description of either public or private keys but not both. It is important when referencing the use of public and private keys for data transmission that candidates also stipulate that it is the recipient's public/private key pairing which is needed to ensure secure data transmission. Candidates should also be reminded of the need to highlight the mathematical relationship between both keys when discussing how they can be used to keep data transfers secure.

Assessment Unit A2 1 Information Systems

Q1 **(a)** Of the three network resources, repeater was the best known with many candidates providing complete or almost complete description of its purpose although inappropriate language such as signals being lost or damaged was common. Many candidates found it difficult to describe the purpose of a server using appropriate DT terminology. This was also the case with a network card, where a significant number of candidates described a MAC address.

(b) P2P networks were appropriately familiar to the majority of candidates and many complete answers were provided.

(c) The overall level of response to this question was disappointing. Many candidates simply referred to user names and passwords without explaining how they contributed to data security.

(d) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate bus networks and star networks with regard to data security, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. Most candidates did not properly describe each topology. Many discussed how an additional node could be added to each, or the consequences of cable failure on each. As expected, very few candidates met the strict criteria for a top-band response.

Q2 **(a)** **(i)** This question required candidates to focus on the purpose of each of the two sets but this proved to be very difficult or impossible for the majority.

(ii) The overall standard of response to this question was high. Most candidates could provide two relevant points for each layer.

(iii) The majority of candidates struggled to relate the coherent layered structure of the OS model to the design of hardware and software components and the subsequent impact on user choice.

(b) Many candidates were properly familiar with both protocol and many correct answers were provided. CSMA/CD was better understood than RFID. In the latter case, many candidates failed to provide appropriate technical detail, many also focused on the advantages of use of RFID compared to bar codes.

(c) This QWC question, purposely designed to evaluate the use of a parity bit in detecting errors, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As in previous series, many candidates struggled to explain the purpose of a parity bit accurately. Some candidates suggested that the choice between odd and even parity and even the whole parity checking process were performed by the user. Others stated that error detection simply involved checking the parity bit after transmission. As expected, very few candidates met the strict criteria for a top-band response.

Q3 **(a)** The overall level of response to this question was high. Most candidates could provide at least three of the required points. Of those candidates who couldn't, a common weakness was the use of non-database terminology. Many candidates did not understand the difference between an entity, an entity occurrence and an attribute.

(b) The overall level of response to this question was very high.

(c) The overall quality of normalization in response to Part (i), Part (ii) and Part (iii) was disappointing. Many candidates did not perform normalization to 1NF correctly and hence could not perform normalization to 2NF and 3NF correctly. It must be stressed again that candidates must understand the importance of annotating their normalization correctly. To get credit, each entity must have an appropriate entity name, the primary key must be identified (by underlining or by a #) and all attributes must be shown.

(iv) As a consequence of incorrect normalization, the overall quality of ER model was disappointing. Again it must be stressed that candidates must use the Entity-relationship diagram notations from the specification.

(d) This question divided the candidature into three almost equal parts. One group were properly familiar with SQL and could accurately produce the required code. A second group were familiar to some extent with SQL and could produce some of the required code, even if inaccurately. The last group appeared to be reduced to guesswork.

Q4 **(a)** Most candidates were properly familiar with the Turing Test, although a significant number failed to provide accurate terminology. A number of candidates ignored the historical context and answered the question with references, often inaccurate, to AI.

(b) The overall level of response to this question was very high and many excellent answers were provided.

(c) The overall level of response to this question was disappointing. Most candidates were not properly familiar with the inference engine.

(d) **(i)** The expert system shell was understood most clearly of the three terms and some excellent descriptions were provided. There was confusion among many candidates about the role of the medical consultants with many believing they created the system or used the system rather than provide the expert knowledge. Even more confusion surrounded the knowledge engineer who was credited with nearly all aspects of the design and use of the system.

(ii) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate the use of expert systems in medicine, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. This is one of the three applications identified in the specification so candidates were expected to have detailed knowledge. Most candidates did not properly describe this application and many evaluations lacked substance. As expected, very few candidates met the strict criteria for a top-band response.

Q5 **(a)** Candidates were more familiar with cells than with handoffs. In fact, only a small number of candidates were properly familiar with the meaning of handoffs.

(b) The overall level of response to this question was very high. The only significant weakness was a confusion in some candidates' minds between robotics and robots.

(c) The overall level of response to this question was disappointing. Most candidates could identify at least one advantage of using robots in this context but there was a general failure to provide the supporting detail required at this level. 'Do not need breaks' and 'are cheaper' were common examples.

(d) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate the use of cloud computing with regard to data security in providing services previously provided by a LAN, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. Only a small number of candidates properly understood cloud computing beyond 'storing data on the web' usually by a single user. As a result, descriptions lacked the required detail and evaluations lacked the required depth and justification. As expected, very few candidates met the strict criteria for a top-band response.

Q6 **(a)** **(i)** In questions about legislation, candidates are required to use the language contained in the relevant legislation as appropriate. The stem of this question deliberately used phrasing from the Computer Misuse Act to remind candidates of this requirement. However, many candidates ignored this with the result that responses referring to 'accessing computers', 'accessing data' or 'for fraud' were common.

(ii) The overall level of response to this question was disappointing. Candidates were required to address the impact of Computer Misuse legislation on an organization. However, many candidates focused purely on individual employees and so could not access many of the acceptable mark points.

(b) **(i)** Only a small number of candidates could accurately state the purpose of the Copyright Designs and Patents Act concisely and accurately as required. Only a small number of candidates referred to 'intellectual property' as was required.

(ii) The overall level of response to this question was disappointing. Candidates were required to discuss the impact of the Copyright Designs and Patents Act on an organization. However, many candidates focused purely on individual employees and so could not access many of the acceptable mark points.

Principal Moderator's Report

Assessment Unit A2 2 Application Development (Case Study)

This is a piece of coursework submitted by Candidates based on a pre-released case study. In this series the case study was called BelFarm and it carries a 20% weighting of the overall A Level outcome. Candidates spend approximately 60 hours documenting and producing a relational database solution for BelFarm. Centres internally assess the candidates work using assessment indicators and submit outcomes to CCEA. Moderation of centre outcomes by CCEA then takes place.

Unit Overview

The pre-released case study for the 2025 series was entitled BelFarm. The Moderator's reported that there was a good variety of coursework solutions submitted from Centres in 2025. Overall the majority of Centres completed the solutions to a good standard and the marks awarded for the most part were accurate and within agreed standards. The samples provided for moderation were of an acceptable standard for A2 Digital Technology. The majority of the samples moderated were presented in appropriate sections following the guidance stated in the specification. This greatly assisted the moderation process. The overall standard of the solutions was good, and many candidates excelled in their application development section, taking into consideration the end user.

There was evidence in the solutions provided that candidates focused on one of the main aspects of the BelFarm case study. Some candidates focused on the booking of activities, some focused on the buying and selling of horses and some focused on an ordering system for Belfarm stock. It was pleasing to see this variety of approaches. There was also evidence in some Centres that candidates continue to produce highly guided solutions which restricted the individual candidates from being able to create their own unique solutions. This should be discouraged as the case study is designed to be undertaken by individual candidates to suit their ability and skills in using Digital Technology to solve problems.

Teacher comments continue to be helpful to the moderation team, in understanding how the teacher has applied the mark scheme. The assessment criteria were applied correctly by the majority of Centres. The quality of documentation and evidence produced by candidates is an area of improvement. Centres are clearly providing clearer instructions and guidance on what candidates need to include in each section and also the standards expected for each of the elements.

The analysis section was approached well by candidates and assessed accurately by Centres. Candidates who were able to clearly identify and document their user requirements went on to produce high quality overall solutions. When only a limited or a brief number of user requirements were identified, candidates found it difficult producing a detailed design section and application development section for their solutions. The higher quality solutions have clear evidence of essential/non-essential requirements with detailed justification included. The quality of DFDs in this section continues to be of a high standard. Candidates should be encouraged to use the dataflow diagram notation as outlined in the specification.

In the design section, the majority of solutions were highly structured in preparation for the application development stage. The overall standard of design work is good, and it is pleasing to see more Centres making better and appropriate use of development tools such as ER diagrams. However, it would be helpful for Centres to encourage candidates to provide the working out of their database normalisation. The better candidates fully explained the different stages of database normalisation as they progressed from 0NF to 3NF using the correct notation. Candidates are also more comfortable using advanced querying functions such as the use of Append and Delete queries alongside Make-Table queries to implement an archive function. The planning of Queries and the use of Pseudocode/ algorithms was more evident, which was pleasing to see. There were a small number of candidates who submitted printscreens of the final forms and reports instead of wireframes. Better candidates submitted designs that could be implemented by a third party.

In the application development section, there has been a noticeable improvement in the quality of user interfaces developed. Many students are also utilising timer objects to implement 'Splash screens'. The standard of relational database applications was generally quite high with many candidates employing a wide range of querying and reporting features in their solutions. Weaker candidates produced solutions that lacked a user friendly interface. For example some solutions provided no navigation back to the original menus. More advanced solutions were user friendly and easy to navigate around all system components. There was a good range of different types of queries created, relating to user requirements set out at the beginning. The Testing section, including the test plan, produced by the majority of candidates covered a wide range of tests in line with user requirements. It was pleasing to see an increase in the inclusion of corrective action documented with printscreens, illustrating fixes as a result of testing.

User documentation was fully understood by centres as to what is expected and the level of detail for each component of the solution. In the evaluation section, most candidates were able to outline the strengths and limitations of their solutions together with ideas for extensions to the system, making reference to essential and non-essential user requirements. Evaluations were comprehensive when they were documented with reference to user requirements.

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